

Serie 7

VECTOR SPACES AND SPAN

1. Linear independence. For each of the following sets, prove whether or not they are linearly independent over $\mathbb{R}$:

$$S_1 = \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \\ -3 \\ 9 \end{pmatrix}, \begin{pmatrix} 1 \\ 3 \\ -4 \\ 7 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \\ 1 \\ 3 \end{pmatrix} \right\},$$

$$S_2 = \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 0 \\ -1 \end{pmatrix} \right\}.$$

2. Linear independence. Are the following sets linearly independent over $\mathbb{R}$?

- (a) $\{(1, 0, 0), (0, 2, t), (2, 4, t^2)\}$ for t in $\mathbb{R}$;
- (b) The set of columns of an upper triangular matrix $A \in M_{n \times n}(\mathbb{R})$ with $A_{ii} \neq 0$ for all $1 \leq i \leq n$. We define an upper triangular matrix to be a matrix whose entries under the diagonal all vanish, i.e. a matrix $A = (A_{ij})_{1 \leq i, j \leq n}$ such that $A_{ij} = 0$ whenever $j < i$.
- (c) $\{f, g\} \subseteq \text{Abb}(\mathbb{R}, \mathbb{R})$, where $f(x) = \sin(x)$ and $g(x) = \cos(x)$;
- (d) $\{f, g\} \subseteq \text{Abb}(\mathbb{R}, \mathbb{R})$, where $f(x) = e^{rx}$ and $g(x) = e^{sx}$, for fixed $s, r \in \mathbb{R}$.

3. Linear independence and span. Consider $A_1, A_2 \in M_{2 \times 3}(\mathbb{R})$, given by

$$A_1 = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

(a) Show that $\{A_1, A_2\}$ is linearly independent over $\mathbb{R}$.

(b) Let

$$M := \left\{ \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} \in M_{2 \times 3}(\mathbb{R}) \mid d = e = 0, b - a = f, 3a = c \right\}$$

Prove that $\text{Sp}(A_1, A_2) = M$.

(c) Find $A_3 \in M_{2 \times 3}(\mathbb{R})$ such that $\{A_1, A_2, A_3\}$ is linearly independent. Is $\text{Sp}(A_1, A_2, A_3) = M_{2 \times 3}(\mathbb{R})$ for any choice of such an A_3 ?

4. Basis of a subspace. Show that

$$U = \{f \in \text{Abb}(\mathbb{F}_5, \mathbb{F}_5) \mid \sum_{i=0}^4 f(\bar{i}) = 0\} \subseteq \text{Abb}(\mathbb{F}_5, \mathbb{F}_5)$$

is a linear subspace. Determine a basis of U .

5. Countable basis. Let V be a vector space over some field K that admits a countable basis. Show that every linearly independent subset $S \subseteq V$ is finite or countable.

6. Linear independence. Prove that the functions

$$\varphi_a : \mathbb{R}_{>0} \rightarrow \mathbb{R}, \quad x \mapsto \frac{1}{a+x}$$

for all $a \in \mathbb{R}_{\geq 0}$ are linearly independent.

Hint: Use that a non-zero polynomial only has finitely many zeros.

Multiple Choice questions. Each question can admit several answers.

Question 1. Let V be a vector space over K . Which of the following assertions is true?

- Let $v \in V$, then the set

$$W := \{w \in V \mid \exists \lambda \in K : w = \lambda v\}$$

is a linear subspace of V .

- A subset $W \subset V$ is a linear subspace if and only if $\text{Sp}(W) = W$.
- Let $S_1, S_2 \subset V$ be subsets. Then $\text{Sp}(S_1 \cup S_2) = \text{Sp}(S_1) + \text{Sp}(S_2)$.
- Let $S_1, S_2 \subset V$ be subsets. Then $\text{Sp}(S_1 \cap S_2) \subseteq \text{Sp}(S_1) \cap \text{Sp}(S_2)$.

Question 2. Let V be a vector space and let $S_1, S_2 \subseteq V$ with $S_1 \subsetneq S_2$. Which of the following are true?

1. If S_1 is a linearly independent set, when is S_2 a linearly independent set?
 - Always
 - Never
 - Sometimes

If now S_2 is a linearly independent set, when is S_1 a linearly independent set?

- Always
 - Never
 - Sometimes
2. Answer the previous question, replacing “linearly independent set” with “generating set for V ”.
 3. Answer question (a), replacing “linearly independent set” with “basis for V ”.